

GENERATED COLUMNS

A **Generated Column** is a column whose value is automatically calculated from other columns in the same row. The database computes their value automatically.

Example:

FirstName	LastName	FullName
Aadya	Ratogi	Aadya Rastogi
Arpit	Raj	Arpit Raj

You only store:

- `FirstName`
- `LastName`

The database computes:

```
FullName = CONCAT(FirstName, ' ', LastName)
```

Types of Generated Columns

Virtual Columns	Stored Columns
• Calculated every time you read the row. • Not stored on disk.	• Calculated once during INSERT/UPDATE. • Stored physically like normal columns.

Syntax

MySQL

```
CREATE TABLE Products(  
    Price DECIMAL(10,2),  
    Quantity INT,  
    Total DECIMAL(10,2)  
    GENERATED ALWAYS AS  
    (Price * Quantity)  
);
```

Whenever `Price` or `Quantity` changes, `Total` changes automatically.

Another Example

```
CREATE TABLE Employees(  
    FirstName VARCHAR(50),  
    LastName VARCHAR(50),  
    FullName VARCHAR(101)  
    GENERATED ALWAYS AS  
    (  
        CONCAT(FirstName, ' ', LastName)  
    )  
);
```

Virtual Generated Columns

These are not stored. Database computes them every SELECT.

Example

```
CREATE TABLE Products(  
    Price DECIMAL(10,2),  
    Quantity INT,  
    Total DECIMAL(10,2)  
    GENERATED ALWAYS AS  
    (  
        Price * Quantity  
    ) VIRTUAL  
);
```

Advantages

- Saves storage
- Always up to date

Disadvantages

- Every SELECT performs the calculation.

Stored Generated Columns

Stored physically.

```
CREATE TABLE Products(  
  Price DECIMAL(10,2),  
  Quantity INT,  
  Total DECIMAL(10,2)  
  GENERATED ALWAYS AS  
  (  
    Price * Quantity  
  ) STORED  
);
```

Advantages

- Very fast SELECT.

Disadvantages

- Uses extra storage.

Rules for Generated Expressions

Generated expressions generally:

- Can reference columns from the same row.
- Cannot reference other tables.

Generated Columns vs Views

This interview question is common.

Generated Column

Inside table

View

Separate virtual table

View Example

```
CREATE VIEW ProductView AS
SELECT
    Price,
    Quantity,
    Price*Quantity AS Total
FROM Products;
```

Generated Column	View
Inside table	Separate object
One table	Multiple tables possible
Automatic per row	Query-based
Can be stored	Usually not stored

Indexing Generated Columns

Suppose `Price * Quantity` is searched frequently.

Without index

Every row



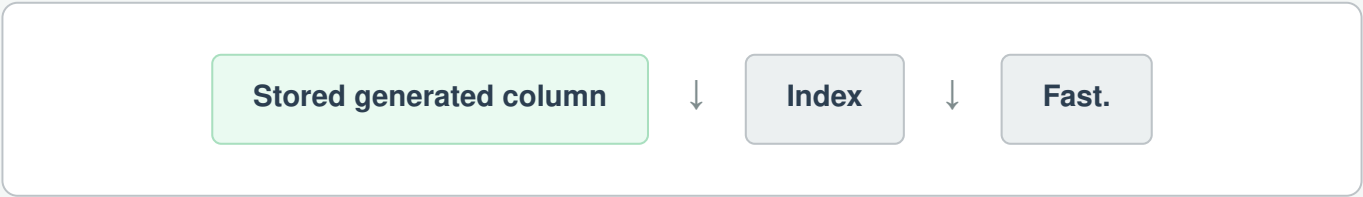
Multiply



Compare

Slow.

Stored generated column



Example

```
CREATE INDEX idx_total
ON Products(Total);
```

In MySQL:

- Stored generated columns can be indexed.
- Virtual generated columns can also be indexed in MySQL 8.0

Vendor Comparison

Database	Support
MySQL	Virtual + Stored
PostgreSQL	Stored only
SQL Server	Computed Columns (virtual or persisted)
Oracle	Virtual Columns
SQLite	Virtual + Stored